



Fig. 1: MDD classification chart for active devices

of the initial discussions and research with doctors.

Then, intellectual property (IP) strategy needs to be decided. And if one is planning to branch out to different countries, IP strategy for each of those countries needs to be planned as well.

Programme plan for the product is also important. The audience should chart out plans for their products around three years in advance.

The alpha and beta prototypes

Once all relevant data has been collected and analysed, prototypes are created. The first is the alpha prototype. This is meant only for testing, and not for the market. Electromagnetic interference and electromag-

netic compatibility tests are also performed at this stage.

Along with the design of the alpha prototype, there are certain regulatory and clinical considerations that need to be kept in mind. There are 14 such notified devices, and Drug Controller General of India (DCGI) needs to be informed about any products related to those devices. Only after approval has been granted, clinical trials can be conducted. Some of these devices are intraocular lenses, orthopaedic implants, heart valves and cardiac stents.

Getting through regulations

Regulations vary from country to country. In Europe, regarding medical devices, there is CE MDD.

CE refers to European Conformity standards for products, and MDD (Medical Devices Directive) refers to standards that are specifically meant for medical products. It deals with three classes of devices that have been classified depending on the amount of risk that the device poses to the patient.

Class I refers to normal medical products that can be disposed of, such as surgical dressings and stethoscopes.

Class II is divided into two categories, A and B. Class IIA consists of medium-risk devices like tracheal tubes. Class IIB consists of higher-risk devices that are surgically-invasive or implantable like intraocular lenses.

Class III devices are the highest-risk implantable devices such as stents and radiology equipment. Based on the results of the tests conducted during the previous stage, the beta prototype is created, which is one step closer to the market.

The USA has certain Food and Drug Administration regulations that involve a similar classification method as CE MDD. It has a category known as 510(k). This is for products that have been developed that are similar to existing devices. The products may have different engineering designs, but these have the same clinical outcomes, so no clinical trials and pre-market surveys are needed.

Essentially, a company needs to show that their device is similar to an existing device, along with the lab test reports of the existing product. Regulatory authorities also provide a detailed framework of rules and regulations, and one should follow the regulations that are applicable to the product. Clinical literature is also important.

Clinical engagement

Another important factor that has to be kept in mind is clinical engagement. This can be done in

Indian case studies

There are many Indian companies that are now shipping medical devices abroad such as Consure, Forus and Perfint. One of the author's first projects was a device that he designed as part of Stanford-India Biodesign Programme.

Funded by Department of Biotechnology, Ministry of Science and Technology, government of India, the programme was centred in New Delhi and administered as a collaboration between Stanford University, USA, Indian Institute of Technology Delhi, All India Institute of Medical Sciences (AIIMS) and Queensland University of Technology. Biotechnology Consortium of India (BCI) manages the IP from the programme.

The product is called Thorashield, and it is designed to remove fluid from the patient's lungs. It is targeted at tuberculosis and cancer patients. Other products that he has been involved in designing include Vapcare, which is designed to reduce the risk of pneumonia in patients who are on the ventilator, and Saans, which is a neonatal CPAP for resource-constrained settings.

